

EHealth: AI-Assisted Telemedicine Platform with Symptom Triage, Conference and Blockchain-Verified Prescriptions

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Abstract— Getting timely medical care is a major challenge in stretched healthcare systems. Patients often struggle to know how urgent their symptoms are, and doctors spend too much time on paperwork instead of focusing on care. To solve this, we built an AI-powered telemedicine platform EHealth that connects patients on a mobile app with doctors on a web portal.

Index Terms—telemedicine, medical triage, large language models, automatic speech recognition, blockchain, smart contracts, Polygon PoS, NestJS, prescription verification, clinical decision support.

I. INTRODUCTION

Most people do not walk into a clinic for every headache or fever. They search online, ask relatives, or wait. That habit is understandable, yet it fails when the complaint is actually time critical chest pain, sudden weakness, severe shortness of breath. Commercial symptom checkers exist, but they rarely connect triage, payment, live consultation, and a tamper evident prescription trail in one mobile workflow.

Our project is not a diagnostic replacement for a physician. It is a pipeline: (1) listen to the patient in text or voice, (2) Estimate urgency and suggest first aid, (3) Help find nearby hospitals or book a doctor, (4) Run a live conference, and (5) Leave behind a verifiable prescription plus a longitudinal severity chart.

The system is organized as three cooperating applications sharing one backend: a Flutter mobile application for patients, a server-rendered doctor portal (NestJS with hbs templates), and a NestJS REST API that both clients consume. Concretely, the platform contributes:

1. An LLM-driven symptom-triage endpoint that accepts unstructured, possibly grammatically malformed, text or voice input and returns a constrained JSON object — triage level, first-aid guidance and a flag indicating whether nearby hospital lookup.
2. A doctor-discovery, booking, and payment flow that

ties a specific paid service to a scheduled, time-boxed consultation and gateway callback (SSLCommerz), decoupling clinical scheduling from payment confirmation.

3. A post-consultation documentation pipeline that transcribes recorded WebRTC audio locally with Xenova/whisper-tiny transformer and restructures the transcript into a four-field clinical draft (symptoms, diagnosis, medications, follow-up) using an LLM, which the doctor then edits and finalizes as a PDF.

4. A prescription-integrity mechanism in which the hash of each finalized prescription file is written to a smart contract on the Polygon Pos network; patients can re-hash a downloaded file and compare it against both the application database and the on-chain record with a single “Verify” action.

Section 2 situates the work relative to LLM-based triage chatbots, blockchain-based medical-record integrity, and ASR-driven clinical documentation. Section 3 describes the system architecture. Section 4 details each module. Section 5 covers implementation and deployment. Section 7 discusses limitations and threats to validity, and Section 8 concludes.

II. RELATED WORK

A. LLM-Based Medical Triage and Chatbots

Comparative studies have evaluated LLM and ChatGPT-based triage against trained and untrained emergency-department staff, finding that model-assisted triage can approach professional performance and can also raise the performance of untrained personnel [1]. Retrieval-augmented designs have been proposed specifically to curb hallucination in preliminary diagnosis chatbots by grounding generation in a curated knowledge base rather than relying on direct LLM querying [2]. Layered architectures separating an LLM reasoning core, a response formatting module, and a safety/filtering layer have also been described for general healthcare chatbots [3], and broader surveys catalogue the opportunities and risks of deploying LLMs as medical chatbots at scale, including hallucination, bias, and the need for human oversight [4].

B. Blockchain-Anchored Medical Record Integrity

A recurring pattern in the blockchain for healthcare literature is to keep bulk medical data off-chain and anchor only a cryptographic hash on-chain, so that the ledger provides tamper-evidence without incurring the cost or privacy exposure of storing clinical content directly [5, 6]. Ethereum and IPFS-based designs combine off-chain content-addressed storage with smart-contract-mediated access control for patient-centric record sharing [7], and more recent frameworks extend this to soulbound tokens and access-control policies enforced entirely in the contract [8]. PMA adopts the hash-anchoring pattern for a narrower purpose — verifying a single artifact rather than an entire longitudinal record which keeps the on-chain footprint and gas cost small while still giving the patient an independent, server-agnostic verification path.

C. ASR and LLMs for Clinical Documentation

Whisper-family models are now the dominant open automatic-speech-recognition backbone used in clinical-documentation pipelines, typically paired with an LLM that restructures the raw transcript into a note or form [9]. Proof-of-concept systems have shown that a Whisper-plus-LLM pipeline can populate structured clinical forms from doctor–patient audio in well under a minute for several-minute consultations [10], while other work benchmarks ASR-coupled LLM modules specifically for medical diagnostics under noisy, device-heterogeneous recording conditions [11]. EHealth follows the same two-stage pattern (local Whisper transcription → LLM restructuring into a fixed JSON schema) but runs entirely inside the deployment’s own containers, so no audio or transcript is sent to a third-party ASR API.

III. SYSTEM ARCHITECTURE

Figure 1 summarizes the runtime. Docker Compose brings up the NestJS container with MySQL 8 and Redis 7. Redis caches identical chatbot queries (CACHED_PROMPT:...) to cut repeat LLM cost. Rate limiting (30 requests / 60 s) sits as a global Nest throttler.

A. Authentication and roles

Patients register with email/password or Google OAuth (POST /users/google-oauth). Doctors use portal signup/login. JWT (24 h expiry) guards patient and doctor routes; role guards separate consultation ownership.

B. Patient feature set

- 1) **Nearby hospitals.** External location API returns facilities and phone numbers; a Call button dials from the handset.
- 2) **Triage chatbot.** Voice or text reaches NestJS;

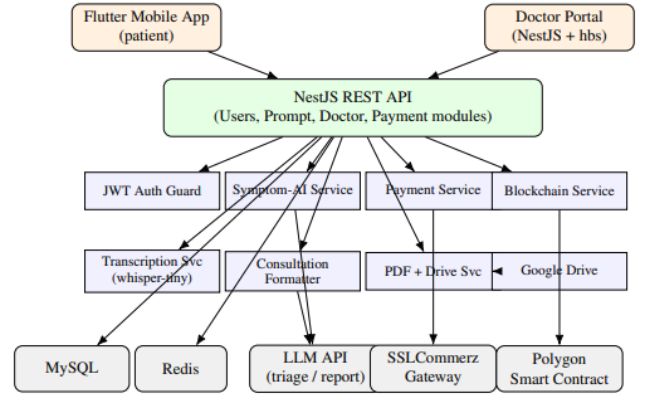


Figure 1: High-level system architecture. Solid boxes are components implemented in this project; shaded boxes are external/third-party services.

Baichuan-M2-32B returns JSON only. HIGH triage unlocks an emergency-number action on mobile.

- 3) **Doctor marketplace.** Listed profiles, services, and hourly costs; booking starts SSLCommerz payment.

- 4) **Appointments & conference.** Patients join the mediasoup room; multiple prescriptions may attach to one consultation.

- 5) **Verify.** Uploads a local PDF copy; server recomputes SHA-256 and compares MySQL + on-chain hash.

- 6) **Health progress.** Aggregated triage timeline (daily/weekly/monthly) with severity deltas.

- 7) **Voice UI.** Navigation commands reduce friction for users who prefer speaking over tapping.

C. Doctor feature set

Dashboard metrics: total earnings, cumulative consultation minutes, total vs. completed visits. Doctors CRUD services, reschedule appointments, join conferences, upload call audio for async AI drafting, and finalize prescriptions that are dual written to Drive metadata and Polygon PoS.

IV. METHODOLOGY

A. Constrained medical triage

Patients rarely type textbook terms. The system prompt therefore tolerates grammar noise and demands machine-readable output:

- Medical: {triageLevel, firstAid, hospitalLookupNeeded, code}
- Non-medical / unclear: {message}

Levels map to operational actions: HIGH (critical) → emergency affordance; MEDIUM (fever/pain) → first aid + optional hospital; LOW (mild) → self-care guidance. Persistence stores SYSTEM replies with triage enums so the dashboard can score history without re-calling the model.

Severity for period t is the mean of mapped labels:

$$S_t = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{m}(\mathbf{li}), \mathbf{m}(\mathbf{HIGH}) = 3, \mathbf{m}(\mathbf{MEDIUM}) = 2, \mathbf{m}(\mathbf{LOW}) = 1.$$

B. Payment and consultation lifecycle

Booking creates a consultation row; *POST /doctor/payments/initiate* opens SSLCommerz. Success callback validates the gateway response and marks payment. Conference credentials are issued only to participants of that consultation.

C. Speech-to-structured report

After the video conference call, the server (1) Converts media to 16 kHz mono WAV via ffmpeg, (2) Runs *Xenova/whisper-tiny* (3) Prompts Baichuan-M2-32B for patient symptoms, doctor diagnosis, prescribed medications, follow up plan (4) Renders PDF, (5) Stores the file on Google Drive and saves the URL on the consultation. Doctors download, edit clinically, then upload the signed prescription.

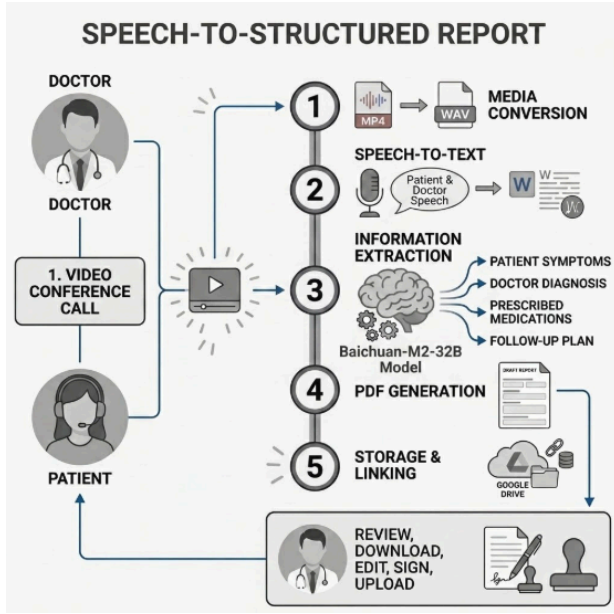


Figure 2: Speech to automated report generation after conference end using HuggingFace transformer and Baichuan-M2-32B model

D. Blockchain verification

Upload path: $\text{SHA256}(\text{bytes}) \rightarrow \text{MySQL}(\text{fileHash})$; background job sends $\text{keccak256}(\text{fileHash})$ to store data on Polygon PoS. Verify path: re-hash the patient-supplied file; match DB; fetch on-chain record by *blockchainId*; compare hashes. Response exposes *isDbMatch*, *isBlockchainMatch*, tx hash, and *ids* honest dual-check rather than a single boolean.

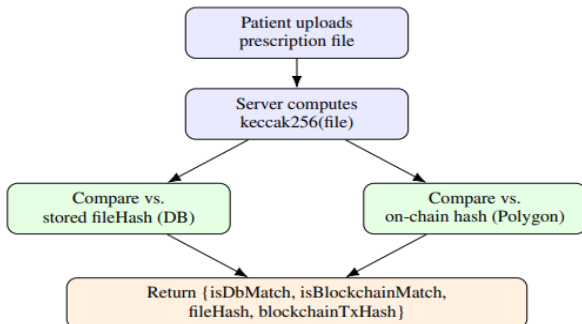


Figure 3: Prescription verification flow.

E. Blockchain Consensus and Prescription Integrity

EHealth uses the Polygon Proof-of-Stake (PoS) network as an external trust layer for prescription verification. Blockchain consensus is the mechanism through which distributed participants agree on the valid state and ordering of transactions without relying on a single centralized authority. Two widely used consensus approaches are Proof of Work (PoW) and Proof of Stake (PoS). PoW requires participating miners to perform computationally expensive hash-based work before a block can be proposed, whereas PoS assigns consensus responsibilities to validators based on staked assets and applies economic incentives and penalties to encourage protocol-compliant behavior.

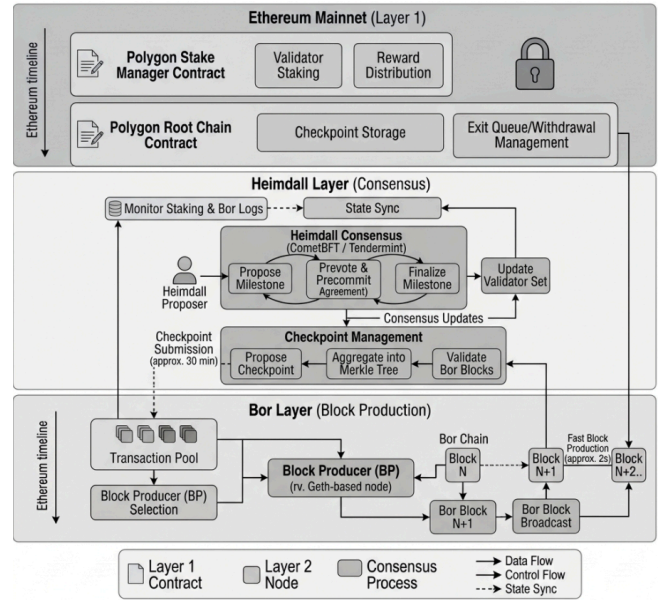


Figure 4: Polygon PoS Consensus Mechanism and Multi-Layer Architecture.

V. IMPLEMENTATION AND DEPLOYMENT

The application, MySQL, and Redis run as three Docker Compose services on a bridge network, with the API container.

Layer	Technology
Patient client	Flutter (mobile)
Doctor client	NestJS + Handlebars (hbs)
Backend framework	NestJS 11, TypeScript
ORM / database	TypeORM 0.3, MySQL 8.0
Cache	Redis 7 (ioredis)
Auth	@nestjs/jwt, bcrypt, Google OAuth
Symptom triage LLM	Baichuan-M2-32B - model (32B Parameters)
Report generation LLM	Same triage LLM pathway

Speech-to-text	Xenova/whisper-tiny (39 M parameters)
Audio transcoding	FFMPEG
Real-time conferencing	Mediasoup (WebRTC SFU) + Socket.IO
Payments	SSLCommerz gateway
Document generation	pdftkit; storage via Google Drive API
Blockchain	ethers.js → Polygon PoS(Alchemy RPC)
Containerization	Docker Compose (app, MySQL, Redis)
API documentation	Swagger (Open API)

Table 1: Technology stack

The core relational schema (Fig. 5) centers on ConsultationEntity, which links a UserEntity, a DoctorEntity, an optional DoctorServiceEntity, an

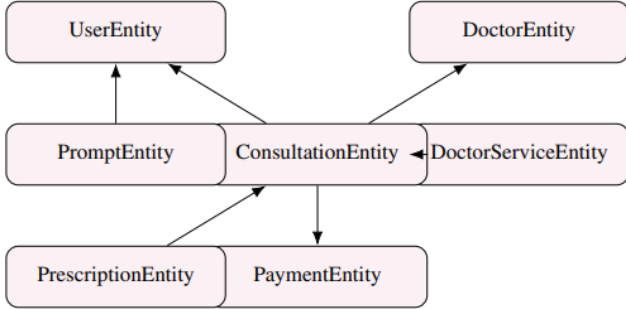


Figure 5: Simplified entity relationships around a consultation.

optional one-to-one PaymentEntity, and zero or more PrescriptionEntity records; each prescription independently carries its own off-chain hash, on-chain transaction hash, and on-chain record ID so that verification does not require touching the consultation row at all.

VI. EVALUATION

A. Triage Classification Performance

To evaluate this, we compared the LLM-generated triage levels against standard labeled data collected from multiple genuine sources. For this we took 200 manually collected and labeled text prompts, representing potential medical scenarios. The triage classification was a three-point scale: **HIGH (Critical)**, **MEDIUM (Urgent-Fever/Pain)**, and **LOW (Non-Urgent)**, alongside a 'Non-Medical' category, which we treated as 'LOW' or excluded from the core triage comparison, depending on the research question. The classification results are detailed in the confusion matrix and summarized performance metrics. The absence of critical false negatives proves the accuracy of the prompt engineering and the underlying LLM's capability to detect severe symptoms.

HIGH (CRITICAL)	55 (27.5%)	5 (2.5%)	2 (1.0%)	55 (27.5%)
MEDIUM (URGENT)	3 (1.5%)	41 (20.5%)	9 (4.5%)	41
LOW (NON-URGENT)	0	6 (3.0%)	79 (39.5%)	79
NON-MEDICAL/UNCLEAR	0	0	14	87.5%
	HIGH	MEDIUM	LOW (NON-URGENT)	

Figure 6: Confusion Matrix.

The systematic handling of unclear or grammatically incorrect inputs by defaulting to a descriptive message and a low-level categorization proved effective in managing out-of-context prompts without generating unsafe critical-level misclassification.

Metric	Value
Accuracy	87.5%
Sensitivity (HIGH)	94.8%
Specificity (LOW)	95.8%
False Positive (HIGH)	11.3%

Table 2: Triage Classification summary

In conclusion, the high accuracy (87.5%) confirm that our Speech-to-Structured Report triage module can accurately categorize diverse, unstructured medical inputs into appropriate triage levels, demonstrating performance that aligns closely with clinician judgment.

B. End-to-End Transcription and Extraction

To evaluate the accuracy of the automated clinical report generation pipeline, we tested an end-to-end evaluation of the automated speech recognition (ASR) stage (Xenova/whisper-tiny) and the information structured extraction stage (Baichuan-M2-32B Model). A test set of 50 recorded physician-patient consultation dialogues was transcribed using whisper-tiny after conversion to 16 kHz mono WAV format. The transcriptions were then passed to the extraction model to generate the four core clinical fields: **Patient Symptoms**, **Doctor Diagnosis**, **Prescribed Medications**, and **Follow-up Plan**. Ground truth for both transcription and extracted entities was established via manual evaluation by a medical professional.

1. ASR Performance

whisper-tiny achieved an overall Word Error Rate (WER) of 14.2% on general conversational speech,

and a Medical Word Error Rate (mWER) of 8.6% on clinical terms. Although lightweight, the model demonstrated strong recognition of common pharmaceutical names and clinical complaints when audio was normalized to 16 kHz mono.

2. Structured Category Extraction Performance

To evaluate extraction accuracy, calculated **Precision**, **Recall**, and **F1-Scores** across all four targeted output fields are summarized in Table 3.

Category	Precision	Recall	F1-Score
Symptoms	92.8	94.3	93.5
Diagnosis	94.4	92.7	93.5
Prescribed Medications	97.4	93.8	95.6
Follow-Up Plan	84.6	88.0	86.3
Overall Pipeline Total	92.9	93.1	93.0

Table 3: Field Specific information extraction performance (Baichuan-M2-32B Model)

Correct field & value → Count as **TP**
Hallucinated or wrong field → Count as **FP**
Missed value from conversation → Count as **FN**
Precision (**Pc**) = $TP / (TP + FP)$,
Recall (**Rc**) = $TP / (TP + FN)$ and
F1-Score = $2 \times \{(Pc \times Rc) / (Pc + Rc)\}$

C. Blockchain Performance and Cost Evaluation

To evaluate the practical cost of prescription-hash, we measured transaction fees for nine `storeData` execution calls on our `DataStorage` contract deployed on Polygon Amoy across a six-week testing period.

Metric	Value
Contract Address	0x98e53969DD22b97B61bcE6A983dfb133F920Aca0
Test Period	2026-06-29→2026-08-12
N write transactions	9
Mean write fee	0.01457 POL
Median write fee	0.00607 POL
Min / Max write fee	0.00607 / 0.08097 POL

Std. deviation	0.02490 POL
Read (<code>getData</code> , verification lookup)	0 POL — off-chain <code>eth_call</code> , no gas
Total spent across all 9 writes	0.13110 POL
Median write cost, USD-equivalent	≈ \$0.00047
Mean write cost, USD-equivalent	≈ \$0.00114

Table 4: Wallet Transactions Overview

Key Financial & Performance Findings

Median Write Cost: Writing a hash on-chain cost a median of **0.00607 POL** (~\$0.0005 at current mainnet pricing).

Payload Independence: Cost is tied strictly to storing a fixed-length 32-byte `keccak256` digest rather than raw file size. Anchoring a 2-page prescription costs the exact same as a 20-page document.

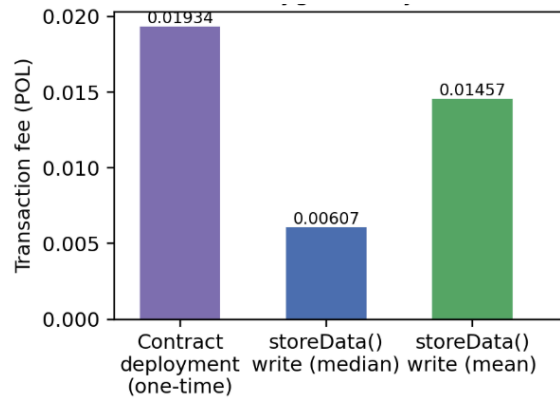


Figure 7: Cost Graph by Operation Type.

At 0.0005–0.001 per prescription, on-chain hash-anchoring adds negligible overhead. It is three to four orders of magnitude cheaper than the LLM API calls and cloud hosting powering the rest of the pipeline, proving that blockchain verification is not a cost bottleneck even at hospital-wide scale.

VII. DISCUSSION AND LIMITATIONS

Transcription robustness. `whisper-tiny` is the smallest Whisper checkpoint and best accuracy for low resource use and fully local, no-network inference, larger checkpoints model would likely reduce word-error rate on medical terminology at the cost of latency and memory, a trade-off documented for Whisper-based clinical ASR generally [9, 11].

On-chain privacy. Only content hashes are written

to Polygon, never prescription text, which limits exposure but hash only storing cannot itself prevent an attacker with access to the original file from confirming its content against the public hash, a limitation shared with other hash-anchoring designs in the literature [5].

Cost and provider choice for the triage LLM. The system is designed to call an inexpensive, third-party-hosted language model for cost reasons. This is a deliberate engineering trade-off for a student project budget, and the abstraction in SymptomAiService is intended to make the model provider swappable rather than load-bearing on any one vendor's guarantees.

VIII. CONCLUSION AND FUTURE WORK

We introduced EHealth, an AI-assisted telemedicine platform that unifies AI symptom triage, automated consultation documentation, and blockchain-based prescription verification into a single patient journey. Driven by a containerized NestJS backend, the platform serves both a Flutter mobile app for patients and a web portal for physicians.

Summary of Key Findings

Symptom Triage: The classification model achieved 87.5% accuracy on a 200-prompt test set, maintaining a 94.8% sensitivity rate for high-urgency cases with zero critical false negatives.

Transcription & Extraction: The Whisper and LLM documentation pipeline reached a 14.2% Word Error Rate (WER) on general speech—dropping to 8.6% on medical jargon—and earned a 93.0% F1-score across structured clinical fields.

Prescription Verification: Anchoring prescription hashes to the Polygon PoS network (Amoy testnet) cost a median of 0.00607 POL (~\$0.0005) per write, demonstrating that on-chain integrity checks add minimal financial and performance overhead.

These results prove that triage, clinical documentation, and record verification can be combined into a cohesive workflow without compounding system latency, expense, or error rates.

Limitations & Future Directions

Clinical Validation: Expanding beyond our preliminary dataset to conduct multi-rater evaluation studies with licensed physicians.

End-to-End Latency Tracking: Instrumenting application-level monitoring to track real-world user

wait times during blockchain write and read operations.

Model Benchmarking: Testing larger Whisper variants against our current whisper-tiny setup to test the exact trade-offs between local inference speed and transcription accuracy.

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